Tiny Think: Reasoning-First Post-Training for Tiny Math and STEM Language Models

Bojan Jakimovski
Faculty of Electrical Engineering & Information Technologies
Ss. Cyril and Methodius University
Skopje, North Macedonia
jakimovski_bojan@outlook.com

Bojan Ilijoski
Faculty of Computer Science and Engineering
Ss. Cyril and Methodius University
Skopje, North Macedonia
bojan.ilijoski@finki.ukim.mk

***Abstract*—We present Tiny Think, an empirical study of a two-stage post-training recipe for 140 million parameter language models designed for single-GPU training environments. Our approach combines supervised fine-tuning on 60 million tokens of math and STEM data with chain-of-thought traces, followed by lightweight preference optimization on 10 million tokens. The key finding is that preference optimization at this scale functions more as a calibration mechanism than as a capability amplifier: while it improves GSM8K accuracy to 9.40 percent, it simultaneously degrades broad reasoning on BBH by 10.66 percentage points and instruction following on IFEval by 5.18 percentage points. We term this tradeoff the "general reasoning tax" and characterize it through ablations across loss objectives, learning rates, and preference hyperparameters. Under the reported GSM8K results considered in this paper, our best model is competitive with and in some cases exceeds the sub-300M baselines included in the comparison, while still exposing substantial regressions that would be missed by math-only evaluation. All artifacts are publicly released for reproducibility.**

***Index Terms*—small language models, supervised fine-tuning, preference optimization, DPO, APO, math reasoning, chain-of-thought**

I. Introduction

Sub-billion parameter language models have emerged as a critical research direction due to their capacity for enabling low-latency inference and practical on-device deployment across resource-constrained environments [1]. However, developing robust reasoning capabilities within such architecturally constrained models remains a significant open challenge in the natural language processing community. Prior research has demonstrated that meaningful reasoning abilities can emerge in tiny models through carefully designed post-training procedures [1], [2], yet the complex interactions between different training objectives, data mixture compositions, and hyperparameter configurations at this parameter scale remain poorly characterized and insufficiently understood.

This paper presents a systematic empirical investigation of *reasoning-first* post-training at 140 million parameters under strict resource constraints utilizing only a single consumer-grade GPU. Our proposed two-stage training recipe consists of the following components: (i) supervised fine-tuning on a carefully curated mathematical and STEM-focused mixture incorporating explicit chain-of-thought reasoning traces totaling approximately 60 million tokens, and (ii) lightweight preference optimization conceptualized as *calibration* rather than reasoning capability amplification utilizing approximately 10 million tokens of preference data. The focus of the paper is the methodology of tiny-model post-training rather than algorithmic novelty: we study how established post-training components interact, how they should be interpreted at this scale, and which tradeoffs they induce in the tiny-model regime.

Key Contribution. Through extensive experimentation, we identify and characterize a phenomenon we term the "general reasoning tax": preference optimization techniques can successfully boost domain-specific mathematical accuracy (GSM8K $+1.36$ pp) while simultaneously inducing substantial degradation in broad reasoning capabilities (BBH -10.66 pp) and instruction-following performance (IFEval -5.18 pp). We present this as an empirical finding about tiny-model post-training behavior rather than as evidence of a universally superior recipe, and argue that multi-objective evaluation is essential when applying preference-based methods at constrained parameter scales.

Comparison with MobileLLM-R1. The MobileLLM-R1 approach employs a two-stage SFT pipeline consisting of general instruction-alignment fine-tuning followed by reasoning-specific fine-tuning utilizing millions of training examples [1]. In contrast, our methodology directly applies a single reasoning-focused SFT stage exclusively on a math+STEM corpus and attains competitive task-specific results with substantially reduced data requirements and computational overhead. This streamlined approach makes the overall training recipe considerably more practical and accessible for academic laboratories and independent researchers operating under resource constraints, although direct math leaderboard comparisons remain sensitive to evaluation protocol differences.

II. Related Work

A. Small Language Models

Recent work has challenged the assumption that reasoning capabilities require large-scale models. MobileLLM [10] introduced architectural optimizations including deep-and-thin designs and embedding sharing for sub-billion parameter models. MobileLLM-R1 [1] extended this with a comprehensive post-training recipe achieving strong reasoning performance at

140M–950M parameters. SmolLM2 [8] demonstrated that data-centric training approaches can yield capable small models, while Gemma-3 [9] showed that architectural innovations enable competitive performance even at 270M parameters.

B. Post-Training for Reasoning

The OLMo-3 "Think" paradigm [2] established a two-stage approach: learning long-form reasoning traces via SFT, then applying preference optimization on paired solutions. Direct Preference Optimization (DPO) [4] provides a simpler alternative to RLHF by directly optimizing on preference pairs. Anchored Preference Optimization (APO) [5] addresses underspecification in alignment through contrastive revisions. Dynamic Fine-Tuning (DFT) [6] reweights token losses based on model confidence to improve generalization.

C. Evaluation Benchmarks

We evaluate on established benchmarks used across the small model literature to ensure comparability with prior work. GSM8K [3] and MATH500 assess mathematical reasoning and are standard benchmarks for MobileLLM-R1, SmolLM2, and Gemma models. BBH [21] (BIG-Bench Hard) evaluates broad reasoning across diverse challenging tasks that require multi-step inference. IFEval [22] measures instruction-following capabilities through verifiable constraints. For STEM evaluation, we use MMLU-STEM [16], ARC-Challenge [17], OpenBookQA [18], and GPQA Diamond [19]. PIQA [20] assesses physical commonsense reasoning. These benchmarks collectively provide comprehensive coverage of mathematical, scientific, and general reasoning capabilities.

III. Dataset Construction

The construction of high-quality training data represents a critical component of our post-training pipeline. Table I provides a comprehensive summary of the dataset statistics employed in our experiments for both training stages.

TABLE I
DATASET STATISTICS FOR SFT AND PREFERENCE OPTIMIZATION STAGES.

Training Stage	Examples	Total Tokens
Supervised Fine-Tuning	29,149	59,999,048
Preference Optimization	2,861	9,999,795

TABLE II
EXAMPLE FROM THE SFT DATASET SHOWING CHAIN-OF-THOUGHT FORMAT WITH EXPLICIT THINK TAGS FOR REASONING TRACE GENERATION.

Prompt: Thomas is training at the gym to prepare for a competition. He trained for 5 hours every day for a month (30 days). If he continues to train for the next 12 days, how many hours will he spend on training in total?

Response: `<think>`In total Thomas would train on $30 + 12 = 42$ days. Thomas trained 5 hours every day, which would bring us to $42 \times 5 = 210$ hours of training in total.`</think>` Answer is 210

A. Supervised Fine-Tuning Mixture

The SFT dataset is constructed by drawing from `allenai/Dolci-Think-SFT-7B` [2] with additional incorporation of GSM8K think-style supervision examples [3]. All training examples are formatted using a standardized chat-style message structure with explicit `<think>` reasoning trace delimiters, as illustrated in Table II. This format encourages the model to externalize its reasoning process before producing a final answer, following the chain-of-thought paradigm that has proven effective for mathematical reasoning tasks. Each data row additionally includes categorical `dataset_source` metadata tags and `token_count` fields, enabling comprehensive post-hoc analysis of source mixture distributions and sequence-length characteristics across the training corpus.

B. Preference Optimization Mixture

The preference dataset is derived from `allenai/Dolci-Think-DPO-7B` [2] and follows a standard preference learning schema comprising `prompt`, `chosen`, and `rejected` response fields, supplemented with `dataset_source` and `token_count` metadata. In this paper, we treat the pair construction procedure as inherited from the source dataset rather than as a new data-generation contribution of our own. According to the source-dataset description in OLMo 3, the `chosen` responses are synthesized with Qwen3-32B in thinking mode and the `rejected` responses with Qwen3-0.6B in thinking mode, so the preference pairs are designed to encode a deliberate capability gap between a stronger and weaker generator [2]. Following that source-dataset setup and the broader delta-learning framing [25], our use of this corpus assumes that the relative quality gap between responses is the key signal for Stage B calibration.

The dataset construction employs a fixed preference token budget of approximately 10 million tokens specifically for the mathematical and STEM reasoning slice. At 140 million parameters, this intentionally constrained data budget is designed to expose sensitivity characteristics to the β regularization parameter and learning rate configurations during Stage B optimization, allowing us to systematically characterize the brittleness of preference optimization under resource constraints.

IV. Methodology

A. Hardware Constraints and Experimental Design

All training experiments are conducted on a single consumer-grade computing platform consisting of one NVIDIA GeForce RTX 5060 Ti graphics processing unit with 16 GB of video memory, an AMD Ryzen 7 9700X central processing unit, and 32 GB of system RAM. We deliberately eschew distributed training frameworks, parameter-efficient fine-tuning adapters such as LoRA [24], and multi-GPU parallelization strategies in order to establish a baseline characterizing what is achievable under strict resource constraints. This experimental configuration represents an accessible research environment readily available to academic laboratories and independent researchers without access to large-scale computational infrastructure.

B. Stage A: Reasoning-Focused Supervised Fine-Tuning

We conduct fine-tuning of the `facebook/MobileLLM-R1-140M-base` foundation model with a maximum sequence length of 4096 tokens. Training employs BF16 mixed-precision computation with gradient checkpointing enabled to fit within the 16 GB VRAM constraint. We apply assistant-only loss computation by masking user conversation turns and incorporate NEFTune embedding noise regularization with α=0.02 [7], which has been shown to improve instruction-following capabilities.

We compare two distinct loss objective formulations. Let π_θ denote the language model parameterized by θ and let $\mathcal{D}$ represent the SFT training dataset:

$$\mathcal{L}_{\text{NLL}} = -\mathbb{E}_{(x,y^\star)\sim\mathcal{D}} \sum_{t=1}^{|y^\star|} \log \pi_\theta(y_t^\star \mid y_{<t}^\star, x) \tag{1}$$

$$\mathcal{L}_{\text{DFT}} = -\mathbb{E}_{(x,y^\star)\sim\mathcal{D}} \sum_{t=1}^{|y^\star|} \text{sg}(\pi_\theta(y_t^\star)) \log \pi_\theta(y_t^\star \mid y_{<t}^\star, x) \tag{2}$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator that prevents gradient flow through its argument [6]. The DFT objective dynamically reweights token losses based on model confidence, potentially improving generalization.

We conduct ablation studies across two key design dimensions: (i) loss objective selection comparing standard negative log-likelihood (NLL) versus dynamic fine-tuning (DFT) [6], and (ii) learning rate magnitude sweeping over $\{2\times10^{-5}, 5\times10^{-5}, 8\times10^{-5}\}$. This range follows established practice for SFT, which typically uses 1–2 orders of magnitude lower learning rates than pretraining [1]. MobileLLM-R1 pretraining employs peak learning rates around 10^{-3}, while their post-training SFT uses substantially lower rates with linear decay to zero. All experimental runs employ an effective batch size of 8 (per-device batch size of 2 with gradient accumulation factor of 4), the Adam optimizer with momentum parameters β=$(0.9, 0.95)$, a cosine learning rate schedule with 10% linear warmup, and gradient norm clipping at 1.0.

C. Stage B: Preference-Based Optimization

We conceptualize preference optimization as a *calibration* procedure, thus improving the model's solution selection capabilities without fundamentally adding new reasoning capacity. We experiment with two preference optimization algorithms: Direct Preference Optimization (DPO) [4] employing a sigmoid-based loss formulation, and Anchored Preference Optimization zero-variant (APO-zero) [5]. We conduct hyperparameter sweeps across $\beta \in \{0.3, 0.5, 1.0, 2.0\}$ controlling the KL divergence penalty strength, and learning rate $\in \{1\times10^{-6}, 2\times10^{-6}, 3\times10^{-6}, 5\times10^{-6}\}$. The preference learning rates are further reduced by an order of magnitude from SFT, following standard practice to avoid catastrophic forgetting of capabilities learned during supervised training.

All preference optimization runs train for exactly 1 epoch with maximum sequence length of 4096 tokens (partitioned as 2048 tokens for prompts and 2048 tokens for completions). At the 140 million parameter scale with a 10 million token budget, this training stage exhibits substantial variance across hyperparameter configurations; consequently, we employ training-set diagnostic metrics as gating signals prior to conducting computationally expensive downstream evaluation. Table III provides a comprehensive summary of all training hyperparameters employed across both experimental stages.

TABLE III
TRAINING HYPERPARAMETERS FOR BOTH STAGES.

Setting	SFT	Preference
Init model	Base†	SFT checkpoint
Objective	NLL / DFT	DPO / APO-zero
Epochs	2	1
Learning rate	2–8×10^{-5}	1–5×10^{-6}
Max seq. length	4096	4096 (2048/2048)
Batch size (eff.)	8	8
Precision	BF16	BF16
Grad. checkpointing	Yes	Yes
Schedule	Cosine, 10% warmup	Cosine, 10% warmup
Grad. clip	1.0	1.0
NEFTune α	0.02	–
Adam β	(0.9, 0.95)	(0.9, 0.95)

†`facebook/MobileLLM-R1-140M-base`

TABLE IV
SFT ABLATION RESULTS REPORTING PEAK GRADIENT NORM AND MAXIMUM TOKEN ACCURACY ACROSS LOSS OBJECTIVES AND LEARNING RATES.

Loss	Learning Rate	Max Grad	Max Acc
NLL	8×10^{-5}	9.56	0.853
NLL	5×10^{-5}	5.47	0.831
NLL	2×10^{-5}	5.50	0.802
DFT	8×10^{-5}	1.83	0.839
DFT	5×10^{-5}	1.84	0.817
DFT	2×10^{-5}	1.89	0.793

V. Training Dynamics Analysis

We present training dynamics as lightweight diagnostic signals for assessing optimization stability and characterizing hyperparameter sensitivity. These diagnostics serve as training-set indicators and complement comprehensive downstream task evaluation.

A. SFT Optimization Stability: NLL versus DFT

Table IV presents the results of our SFT ablation experiments, revealing that the DFT objective yields substantially lower peak gradient norms compared to the standard NLL objective (maximum of 1.89 versus 9.56) while achieving comparable maximum token prediction accuracies. This observation suggests that DFT provides more stable optimization dynamics at this parameter scale. This stability advantage is particularly relevant for resource-constrained training where gradient explosions can waste limited compute budgets. Figure 1 visualizes the complete gradient norm trajectories throughout training, clearly illustrating the stability advantages of the DFT formulation.

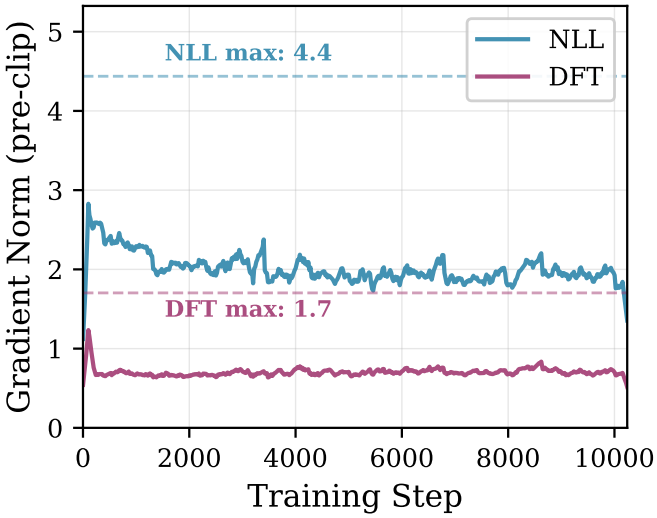

Fig. 1. Gradient norm trajectories comparing NLL versus DFT loss objectives throughout SFT training. DFT consistently maintains substantially lower peak gradient norms than NLL while achieving comparable token prediction accuracies.

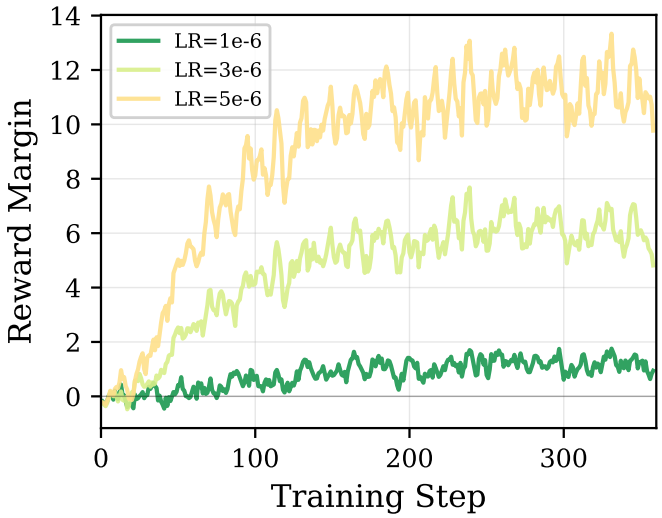

Fig. 2. Reward margin evolution for DPO (β=1) across learning rates. Higher learning rates produce larger margins but do not uniformly improve downstream performance.

TABLE V
PREFERENCE OPTIMIZATION TRAINING DIAGNOSTICS SHOWING MAXIMUM GRADIENT NORM AND FINAL REWARD MARGIN.

Algorithm	β	LR	Max Grad	Margin
DPO	0.5	2×10^{-6}	1776	3.44
DPO	1.0	1×10^{-6}	3968	0.04
DPO	1.0	2×10^{-6}	3920	4.82
DPO	1.0	3×10^{-6}	3856	7.37
DPO	1.0	5×10^{-6}	3856	13.86
DPO	2.0	2×10^{-6}	8384	8.50
APO-zero	0.3	3×10^{-6}	422	3.27
APO-zero	0.5	3×10^{-6}	664	4.58
APO-zero	1.0	3×10^{-6}	1240	6.28

B. Preference Optimization Brittleness

Table V presents diagnostic metrics for preference optimization, revealing high sensitivity to hyperparameter configurations. Higher learning rates consistently yield larger final reward margins between chosen and rejected responses; however, these increased margins do not uniformly translate to improved downstream task performance (as demonstrated in Table X). We therefore treat these large margins and gradient norms as descriptive training diagnostics indicating brittleness and sensitivity, not as mechanistic explanations or stability guarantees. Figure 2 visualizes reward margin evolution across different learning rate settings.

VI. EXPERIMENTAL RESULTS

We conduct all evaluations using the vLLM inference backend with the lm-eval evaluation harness [23] under single-sample stochastic decoding with temperature 0.6 and top-p nucleus sampling at 0.95. These settings are chosen deliberately to match the MobileLLM-R1 reasoning-evaluation protocol [1] and to align with published Qwen3 guidance for thinking-mode inference, which recommends temperature 0.6 and top-p 0.95 and explicitly discourages greedy decoding because it can degrade performance and induce repetitions [11], [12]. We therefore treat sampled decoding here as the standard reasoning-model evaluation setup rather than as an auxiliary ablation variable.

TABLE VI
SELECTED TINY THINK CHECKPOINT RESULTS (ACCURACY %; HIGHER IS BETTER).

Checkpoint	GSM8K	MATH500	BBH	IFEval
SFT (NLL; epoch 2)	8.04	4.00	**23.84**	**21.63**
DPO (β=1, LR 3×10^{-6})	**9.40**	1.80	13.18	16.45
APO (β=0.5, LR 3×10^{-6})	8.26	3.80	12.01	16.08

A. The Math–Reasoning Tradeoff

Table VI presents our central experimental finding. The checkpoint achieving the highest GSM8K accuracy (DPO with β=1, learning rate 3×10^{-6}) attains 9.40% accuracy but critically does not dominate across all evaluation dimensions: BBH broad reasoning drops substantially from 23.84% to 13.18%, and IFEval instruction-following performance decreases from 21.63% to 16.45% relative to the strongest SFT checkpoint configuration.

B. Comparison Against Tiny Model Baselines

Table VII presents a comparative analysis of our best-performing GSM8K checkpoint against reported zero-shot GSM8K results for similarly-sized language models. Within the reported GSM8K results considered in this comparison, our DPO-optimized checkpoint exceeds the listed sub-300M baselines, including Gemma-3-270m-it (8.4% accuracy), while using a substantially smaller post-training budget than the

TABLE VII
GSM8K COMPARISON TO TINY BASELINES (ACCURACY %).

Model	GSM8K
SmolLM2-135M-Instruct [8]	2.4
MobileLLM-R1-140M [1]	4.1
Gemma-3-270m-it [9]	8.4
Tiny Think SFT (NLL)	8.04
Tiny Think DPO	**9.40**

full two-stage MobileLLM-R1 recipe. We therefore interpret Table VII as evidence that the proposed post-training methodology is competitive at very small scale, while reserving broader leaderboard claims because math-evaluation protocols are not fully unified across all sources.

C. STEM and Broader Evaluation

Table VIII shows comprehensive results across the full evaluation suite. Tiny Think checkpoints remain within the same broad STEM performance band as comparable baseline models while exhibiting movement along a math-versus-general capabilities frontier. The DFT objective achieves the highest MMLU-STEM performance (25.88%) but correspondingly lower GSM8K accuracy, further evidencing the tradeoff phenomenon.

D. Incremental Effect of Preference Optimization

To precisely isolate the contribution of the preference optimization stage within our two-stage training pipeline, Table IX reports the performance change from the SFT baseline (NLL; epoch 2) to the best DPO checkpoint as absolute percentage-point deltas. While DPO successfully increases GSM8K accuracy (+1.36 pp) and yields improvements on PIQA (+1.85 pp), the most pronounced performance regression occurs on the BBH broad reasoning benchmark (-10.66 pp), providing clear quantitative evidence for a measurable general-reasoning tax. This pattern of domain-specific gains coupled with broader capability regressions aligns with recent observations in the preference optimization literature suggesting that such methods may inadvertently narrow the model's response distribution [4]. Figure 3 provides a visualization of this tradeoff pattern.

E. Frontier Analysis Across Hyperparameter Sweeps

To comprehensively characterize the capability tradeoff landscape, Table X reports checkpoints spanning our experimental sweep across SFT objectives, learning rates, and preference configurations. The highest GSM8K accuracy values are consistently achieved by DPO settings that simultaneously exhibit substantially lower BBH and IFEval performance compared to the strongest SFT checkpoints, providing robust empirical evidence consistent with our "calibration, not free lunch" conceptual framing.

The choice between NLL and DFT objectives during SFT creates distinct starting points along the capability frontier. DFT checkpoints tend to preserve stronger BBH performance (24.37% for DFT epoch 2 versus 23.84% for NLL epoch 2)

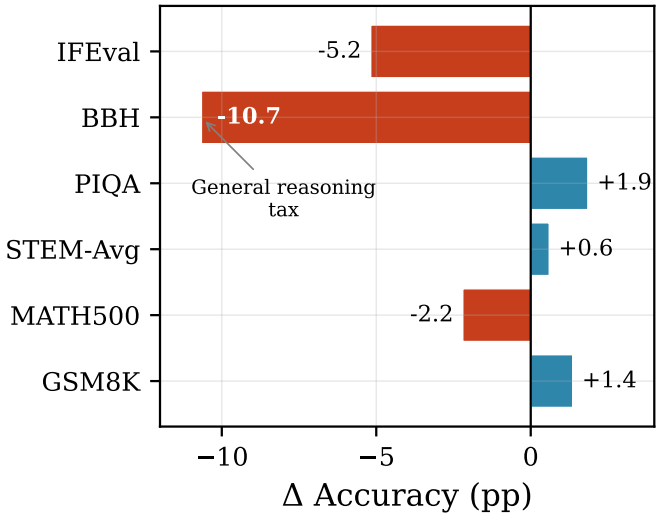

Fig. 3. Incremental effect of preference optimization (DPO − SFT). The substantial BBH regression (-10.66 pp) quantifies the "general reasoning tax."

while achieving slightly lower GSM8K scores, suggesting that the optimization stability benefits of DFT translate into more balanced downstream capabilities.

Figure 4 visualizes DPO performance across the complete $\beta\times$ learning rate hyperparameter grid. The optimal GSM8K performance (9.4% at β=1, LR= 3×10^{-6}) coincides with mid-range BBH accuracy, definitively confirming that the tradeoff phenomenon operates at the hyperparameter selection level.

VII. DISCUSSION

A. The General Reasoning Tax

Our central empirical finding is that preference optimization at the 140 million parameter scale operates fundamentally as a *steering* mechanism rather than a capability amplifier. Table X makes this tradeoff explicit: checkpoint configurations achieving the highest GSM8K mathematical reasoning accuracy simultaneously exhibit substantially degraded BBH performance compared to the strongest SFT checkpoint configurations. This regression is notable because BBH is not a narrow math benchmark but a 23-task suite designed to probe difficult, often multi-step reasoning problems where chain-of-thought prompting is especially informative [21]. In that sense, the contribution of this paper is primarily diagnostic: it maps an empirically observable frontier between math gains and broader CoT-sensitive reasoning regressions under tiny-model constraints.

We hypothesize three contributing factors underlying this phenomenon, but we do not claim to isolate them causally in the current study. In particular, we do not perform task-level BBH analysis or ablations that would let us attribute the regression primarily to one mechanism over the others:

Format Specialization. Preference optimization objectives can sharpen model responses toward highly specific output formats (e.g., verbose step-by-step mathematical derivations

TABLE VIII
COMPREHENSIVE EVALUATION SUITE (ACCURACY %). STEM-AVG IS THE MACRO-AVERAGE OF MMLU-STEM, ARC-C, OBQA, AND GPQA.

Model	MMLU-STEM	ARC-E	ARC-C	OBQA	GPQA	PIQA	BBH	IFEval	STEM-Avg
MobileLLM-R1-140M [1]	23.34	34.13	26.11	33.60	28.79	57.29	18.63	13.68	27.96
Gemma-3-270m-it [9]	21.73	–	26.79	35.40	26.26	–	–	–	27.54
SmolLM2-135M-Instruct [8]	24.01	–	28.75	33.00	21.21	–	–	–	26.74
SFT (NLL; epoch 2)	21.34	34.39	25.85	32.40	22.73	58.87	**23.84**	**21.63**	25.58
SFT (DFT; LR 5×10^{-5})	**25.88**	34.13	23.63	32.60	24.75	59.90	23.78	15.71	**26.72**
DPO (β=1, LR 3×10^{-6})	21.50	34.26	25.09	33.40	24.75	**60.72**	13.18	16.45	26.18

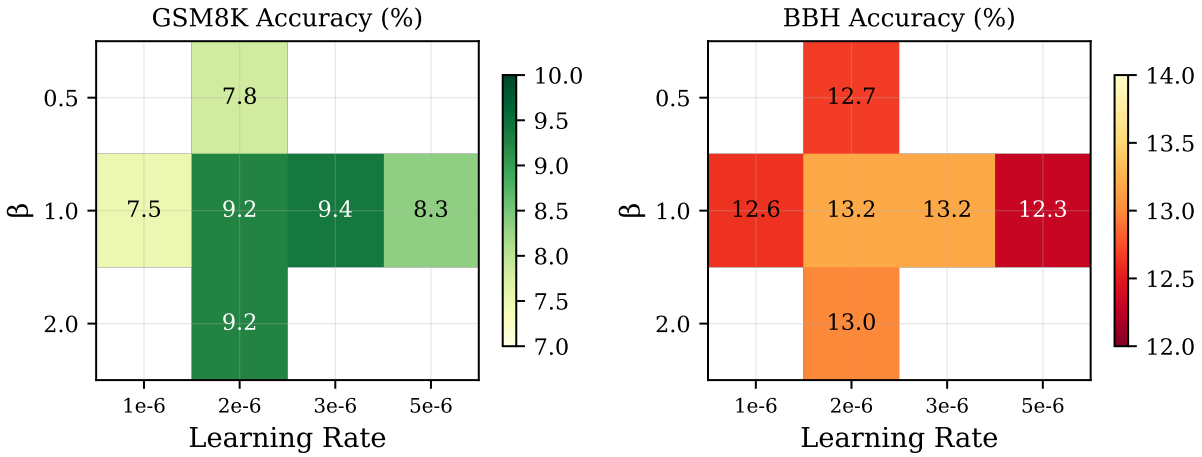

Fig. 4. DPO hyperparameter sensitivity heatmap across $\beta\times$ learning rate. Left: GSM8K accuracy (green = better). Right: BBH accuracy (inverted colormap). Optimal GSM8K (9.4%) at β=1, LR= 3×10^{-6} coincides with mid-range BBH.

TABLE IX
INCREMENTAL EFFECT OF PREFERENCE TUNING (DPO − SFT).

Metric	Δ (pp)
GSM8K	+1.36
MATH500	−2.20
STEM-Avg	+0.60
PIQA	+1.85
BBH	−10.66
IFEval	−5.18

TABLE X
FRONTIER SNAPSHOTS ACROSS SWEEPS (ACCURACY %).

Configuration	GSM8K	BBH	IFEval	STEM-Avg
SFT (NLL; epoch 2)	8.04	23.84	**21.63**	25.58
SFT (NLL; LR 2×10^{-5})	7.58	12.81	16.27	25.97
SFT (DFT; epoch 2)	7.20	**24.37**	18.30	25.80
SFT (DFT; LR 5×10^{-5})	6.75	23.78	15.71	**26.72**
DPO (β=1, LR 1×10^{-6})	7.51	12.62	14.97	25.76
DPO (β=1, LR 2×10^{-6})	9.25	13.19	16.45	25.73
DPO (β=1, LR 3×10^{-6})	**9.40**	13.18	16.45	26.18
DPO (β=1, LR 5×10^{-6})	8.34	12.32	14.60	25.72
APO (β=0.5, LR 3×10^{-6})	8.26	12.01	16.08	25.60
APO (β=1.0, LR 3×10^{-6})	7.96	11.98	16.45	25.83

with boxed final answers) that do not transfer effectively to other task types requiring different response structures.

Capacity Constraints. At 140 million parameters, the model architecture may fundamentally lack sufficient representational capacity to simultaneously maintain strong performance across diverse reasoning task categories when optimization pressure is applied heavily toward a single specialized domain.

Domain Narrowing. The math+STEM trace-heavy composition of our preference dataset may inadvertently overemphasize verbose, highly structured reasoning patterns at the expense of the concise, flexible general-purpose response capabilities required for broader evaluation benchmarks.

B. SFT Objective Selection: NLL versus DFT

The DFT loss objective yields substantially more stable optimization trajectories characterized by lower peak gradient norms compared to standard NLL while achieving comparable token prediction accuracies. However, we observe mixed downstream task results: the strongest BBH checkpoint employs the DFT objective, while the strongest GSM8K checkpoint employs NLL. We therefore do not claim that either objective uniformly dominates; rather, objective selection appears to move checkpoints along the math-versus-general capabilities frontier without strict Pareto improvement.

Interpreting these math results requires scale-appropriate

context. Our goal is not to claim competitiveness with frontier large-model reasoners, but to understand what post-training methodology can recover within the tiny-model regime. In that regime, the relevant comparison is against similarly sized models, and Table VII shows that careful post-training remains meaningful even though the overall capability ceiling at 140 million parameters is still constrained.

C. Limitations

Single-Seed Evaluation Under a Standard Reasoning Protocol. All reported evaluation metrics employ a single random seed under the stochastic decoding configuration used throughout the paper (temperature 0.6, top-p 0.95). We adopt this protocol intentionally because it matches MobileLLM-R1 and published Qwen3 guidance for thinking-mode inference [1], [11], [12]. The limitation is therefore not the use of sampled decoding itself, but the absence of repeated-seed estimates and confidence intervals quantifying sensitivity under this standard reasoning-evaluation setup.

Scoped Objective and Adaptation Coverage. We study DPO and APO-zero as representative preference-optimization objectives within a fixed reasoning-first recipe, rather than attempting an exhaustive benchmark of all recent DPO variants. Alternatives such as ORPO [13] and SimPO [14] are meaningful follow-up comparisons, but our goal here is to characterize whether preference calibration helps or harms tiny reasoning models under a compact, reproducible setup rather than to rank every recent objective variant. Likewise, we deliberately use full fine-tuning rather than LoRA because, at this parameter scale, full fine-tuning is already feasible on a single 16 GB GPU and avoids introducing an additional adapter-capacity variable into the core comparison. Recent work suggests LoRA can be effective in larger small-model or RL-based reasoning settings [15], but we do not view adapter-based comparisons as prerequisite to the present tiny-model methodology study.

Domain Coverage. Our post-training and evaluation focus on math and STEM; broader domains (coding, multilingual, safety) remain uncharacterized.

VIII. Conclusion

We presented *Tiny Think*, a reproducible two-stage reasoning-first post-training recipe for 140 million parameter language models and used it as the basis for an empirical diagnostic study of tiny-model post-training. Through systematic ablation studies spanning SFT loss objectives (NLL vs. DFT), preference optimization algorithms (DPO vs. APO-zero), and hyperparameter configurations, we identified and characterized a "general reasoning tax": preference optimization can improve mathematical reasoning accuracy (GSM8K $+1.36$ pp) while simultaneously degrading broad reasoning capabilities (BBH -10.66 pp) and instruction-following performance (IFEval -5.18 pp).

Our key practical takeaway is that post-training at sub-billion parameter scales should be conceptualized as *capability steering* rather than capability addition. Preference optimization exhibits high variance at this scale and requires explicit regression monitoring across diverse evaluation benchmarks. We publicly release all training curves, model configurations, and evaluation outputs to support reproducibility.

Appendix A
Evaluation Details

We document metric keys from lm-eval: GSM8K uses `exact_match,flexible-extract`; MATH500 uses `math_verify`; ARC-C, OBQA, GPQA, and PIQA use `acc_norm`; BBH uses `exact_match,flexible-extract`; IFEval uses `prompt_level_strict_acc`. All runs use single-sample stochastic decoding (temperature 0.6, top-p 0.95, seed 1234), matching MobileLLM-R1's protocol and Qwen3's recommendations for reasoning models. We do not report repeated-seed confidence estimates or greedy-decoding baselines in the current manuscript.

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